Curriculum Vitae of DR. TARIQ AZIZ

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Associate Professor (Soil Science)

University of Agriculture, Faisalabad

Regional Director (South Asia)

International Nitrogen Initiative (INI) www.initrogen.org

Member Steering Committee

Global Partnership on Nutrient Management (GPNM)

United Nations Environment Programme (UNEP)

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ACADEMIC INFORMATION

Degree/Certificate	Subject	Institution	Year
B.Sc. (Hons.)	Agriculture,	University of Agriculture, Faisalabad	1995-99
M.Sc. (Hons.)	Soil Science	University of Agriculture, Faisalabad	1999-01
Post Graduate Training	Plant Nutrition	of Nutrition, ,	2005
Ph.D.	Soil Science	of ,	2002-06
Post Doc.	Plant Physiology	University of Western Australia, School of Plant Biology	2010-11

PROFESSIONAL EXPERIENCE

Position	Organization	Duration
Principal, Sub-Campus Depalpur,	University of Agriculture, Faisalabad	2018 to 09.10.2023
Project Director	University of Agriculture, Faisalabad, HEC Funded development Project Rs. 889 million	June 2021 to June 2022
Project Director, Sub-Campus Depalpur	University of Agriculture, Faisalabad (Punjab funded development project worth Rs. 453.84 million)	20th January 2016 to Dec 2019
Associate Professor	University of Agriculture, Faisalabad	29th May, 2013 to date
Assistant Professor Tenure Track Basis	of ,	8th October, 2008 to 24th May, 2013
Adjunct Senior Lecturer	University of Western Australia	October, 2014 to date
Adjunct Lecturer	University of Western Australia	Sept 2011 to Sept., 2014
Post Doc Fellow		8th December, 2010 30th Sep, 2011
Assistant Professor	of ,	3.1.2008 to 7th October
Lecturer (regular)	of ,	16.7.2007 to 2.1.2008
Lecturer	of ,	1.12.2006 to 15.7.2007
Teacher per lecture basis	of , , Depalpur Campus	16.10.2006 to 30.11.2006

Research Associate	of	29.3.2003 to 30.7.2004

ACADEMIC AND PROFESSIONAL DISTINCTIONS

1. Best Research Paper Award 5th and 6th HEC Oust standing Research Awards, 2013-14 and 2014-15
2. Research Productivity Award (2012 & 2013) by Pakistan Council of Science & Technology.
3. **Adjunct Faculty at School of Plant Biology**, University of Western Australia. (**2011-2017**)
4. Post Doc Fellowship 2010 at University of Western Australia, Australia, under the supervision of **Prof. Hans Lambers, Head, School of Plant Biology, UWA**.
5. Visiting Scientist at Institute of Plant Nutrition, **JLU, Giessen, Germany** (6 months) from German Exchange Program (DAAD), 2005.
6. Merit Scholarship for Ph.D. studies from Higher Education Commission,

FUNDED RESEARCH PROJECTS/CONSULTANCIES

1. Principal Investigator (Pak Chapter) in **South Asian Nitrogen Hub**, funded by UKRI under GCRF, UK (Total Project Cost: **19.6 Million pounds**) (2019-2023). Pak. Share: 380000/- GBP equivalent to Pak Rs. 100 million).
2. Team member of ICARDA led USDA funded project “Fertilizer Right Pakistan”, PI Dr Aamer Maqsood, worth USD **65000**.
3. Co-PI in “Improvement in citrus productivity and quality by standardization of nutrient assessment protocols for optimization of nutrient applications” by ALP, PARC, amounting Rs. 3.154 million. 2018-20
4. As Team Leader (Pakistan) in a collaborative research project “Sustainable production of bioenergy and soil conditioners from bio residues in Pakistan for energy and food supply” funded by German Government. Total amount: **Euro 100,000**. (2018-19)
5. As PI of research project "Enhancing nutrient use efficiency and fortifying food crops through microbially synthesized nano-fertilizers (bio-nano-fertilizer)” funded by HEC. Total amount Pak. **Rs. 3.65 million**. 2018-21
6. As Principal Investigator of research project “Use of Silicon for amelioration of salinity stress in maize: A sustainable approach” funded by HEC. Total amount: Pak. **Rs. 2.234 million**. 2012-14.
7. As Principal Investigator of research project “Testing of Enhanced Efficiency Fertilizers” funded by Fauji Fertilizer Company (Pvt.) Ltd. MoU signed between UAF and FFC. Pak. Rs. **0.3 million**. (2016)
8. As Co-Principal Investigator of project “Zinc biofortification of wheat grains through farmers’ friendly approaches’ funded by HEC. Total amount: Pak. Rs. 2.34 million. 2011-13.
9. As Co-Principal Investigator of project “Enhancing Bioavailable Zn content in Lentil: A means to higher profits and improved Zn nutrition for rural Punjab” funded by HEC. Rs. 2.3 Million 2018-20.
10. As Co-Principal Investigator “Growth of Bamboo species at different soil salinity levels” funded by HEC. Total Amount: Pak. 0.48 million. 2016.

11. As team Member of Internationally funded project “Nitrogen budgeting in Indus Basin of Pakistan” by ICARDA, Total Amount: US$ 10,000/-. 2017

Consultancies

1. A consultancy project with Engro Fertilizers on Forecasting urea demand for future 2 decades” worth **Rs. 3.0 million**. (July to November, 2025)
2. "Baseline study to identify potential threats to agrobiodiversity and opportunities for its conservation in potato landscape” funded by CABI, 2022. Funds: Rs. 250,000/-
3. “Consultant as Soil Scientist in project “Biodiversity in Potato Landscape” for CABI. Funds, 267,000/-

SUPERVISION OF RESEARCH STUDENTS

M.Sc. (Hons.) Produced:Currently enrolled

Supervisor482

Member of supervisory committee906

Ph.D.

Supervisor50

Member of supervisory committee100

ADDITIONAL / ADMINISTRATIVE ASSIGNMENTS

20.02. 2016 to 09.10.2023	**Principal**, UAF Sub-Campus Depalpur, Okara
2021	**Project Director** HEC funded Project UAF Sub-Campus worth Rs. 889 million. w.e.f 01.08.21 to 2022
2016 to 2019	**Project Director**, UAF Sub-Campus Depalpur, Okara, (20 Jan, 2016) Punjab Funded Project worth Pak Rs 453 million and HEC funded Pak Rs. 889 million.
2019-24	Project Lead partner from Pakistan in "South Asian Nitrogen Hub"
2018-20	Member Discipline Committee, UAF, for two years w.e.f 17.03.2018.
2016	Focal Person for UAF / FAP Collaborative Outreach Activities vide EFS/1538-44, dated 08.11.2016
2013 to 16.11.2016	Deputy Hall Warden, University of Agriculture, Faisalabad.

PUBLICATIONS:

BOOKS

1. Abbas, A., Yang, A., Aziz, T., Wakeel, A. (2024) Pakistan Nitrogen Policy Report: Scientific Evidence, Current Initiatives and Policy Landscape. South Asia Nitrogen Hub (SANH) Policy Paper PP2. Faisalabad.
2. Brownlie, W.J., Aziz, T., Sutton, M.A., …….. and Yan, X., (2024) Nitrogen Mitigation. INMS Guidance Document on Measures for Sustainable Nitrogen Management. INMS Guidance Document Series (Series Editors: M.A. Sutton, M. Schlegel, J. Baron and H.J.M. Van Grinsven). International Nitrogen Management System, UK Centre for Ecology & Hydrology, Edinburgh, UK. ISBN: 978-1-906698-90-4. DOI: 10.5281/zenodo.13831114.
3. Tariq Aziz et al., (Eds) 2021. Nitrogen Assessment; Pakistan as a case study. Elsevier Academic Press Publishers, USA. ISBN: 9780128244173

BOOK CHAPTERS

1. Nitrous oxide scenarios and abatement measures. 2024. CHAPTER 3 in: United Nations Environment Programme and Food and Agriculture Organization. 2024. Global Nitrous Oxide Assessment. Nairobi. https://doi.org/10.59117/20.500.11822/46562. pp.65-86.
2. Awan, M.I, M. I. Sohail, S. Hussain, M.A. Maqsood, M Nasim, and Tariq Aziz. 2024. Nitrogen Use Efficiency and Agricultural Sustainability. In: Rehman, M., J.C. Biswas and R.S. Meena Eds., Climate Change and Soil-Water-Plant Nexus, Agriculture & Environment. Springer Nature, Singapore. https://doi.org/10.1007/978-981-97-6635-2
3. Aziz T., A. Wakeel, A.N. Shahzad, R Rees and M. Sutton. 2021. Rethinking nitrogen use: need to plan beyond present. In: Nitrogen Assessment; Pakistan as a case study. Elsevier Publishers, USA. Pp.1-10
4. Awan, M.I., S. Raza, A Farooq, A. Nawaz and T. Aziz. 2021. Drivers of increased nitrogen use in Pakistan. In: Nitrogen Assessment; Pakistan as a case study. Elsevier Publishers, USA. Pp. 53-68.
5. Singh, B., H. M Bilal and T. Aziz. 2021. Nitrogen use efficiency in crop production: issues and challenges in South Asia. In: Nitrogen Assessment; Pakistan as a case study. Elsevier Publishers, USA. Pp.127-143.
6. Watto, M.A. T. Aziz, A Wakeel, W Ahmad, Abdul Jalil Marwat and Munir Hussain Zia. 2021. Pathways to sustainable nitrogen use and management in Pakistan. In: Nitrogen Assessment; Pakistan as a case study. Elsevier Publishers, pp. 177-189.
7. Maqsood, M.A., **T. Aziz**, M. Sabir, A. Wakeel, M. Sanaullah and M.Z. Rehman. 2017. Soil Organic Matter: Significance, Sources and Functions. Ch. 8. In: Soil Science, concepts and applications. Ed. Sabir et al. Managing Ed. Khan, and Farooq. University of Agriculture, Faisalabad. Pakistan.
8. Wakeel, A., M. Yaseen, M. A. Maqsood, M. Sanaullah and **T. Aziz.** 2017. Plant Nutrients and Soil Fertility Management. Ch. 7. In: Soil Science, concepts and applications. Ed. Sabir et al. Managing Ed. Khan, and Farooq. University of Agriculture, Faisalabad. Pakistan.
9. Khan, W.D., **T. Aziz**, M.A. Maqsood, M. Sabir, H.R. Ahmad, P.M.A. Ramzani and M. Naseem. 2016. Silicon: A beneficial nutrient under salt stress, its uptake mechanism and mode of action. In: Hakeem et al., (Ed.) Soil Science: Agricultural and Environmental Prospective pp 287-301. ISBN: 978-3-319-34449-2 Springer, Switzerland.
10. Ahmad, HR, **Tariq Aziz**, Muhammad Zia-ur-Rehman, Muhammad Sabir, and Hinnan Khalid. Sources and Composition of Waste Water: Threats to Plants and Soil Health. In: Soil Science: Agricultural and Environmental Prospective. 349. ISBN 978-3-319-34449-2 Springer International Publishing Switzerland 2016.
11. **Tariq Aziz**, M. A. Maqsood, S. Kanwal, S. Hussain, H.R. Ahmad and M. Sabir. 2015. Fertilizers and Environment: Issues and Challenges. In: Crop Production and Global Environmental Issues. pp 575-598. Ed. Hakim, K.R. ISBN: 978-3-319-23161-7 (Print) 978-3-319-23162-4 (Online). Springer International Publishing Switzerland
12. Ahmad, H.R., **Tariq Aziz,** M. Z. Rahman and Saifullah. 2015. Spatial mapping of metal contamination, In: Soil Remediation and Plants, Prospects and Challenges Elsevier. pp 415-431. K.R. Hakeem et al. (Ed.) Academic Press, Elsevier. USA. ISBN: 978-0-12-799937-1
13. **Aziz, T.,** M. Sabir, M. Farooq, M.A. Maqsood, H.R. Ahmad. 2014. Phosphorus deficiency in plants; Responses, adaptive mechanisms and signaling. In: Plant Signaling; Understanding the molecular cross-talks, Hakim et al. (Ed.) pp. Springer.
14. Jabran, K., M. Farooq, **Tariq Aziz** and K.H.M. Siddique. 2013. Allelopathy and Crop Nutrition. Ch. 14. In: Z.A. Chema *et al* (eds.) Allelopathy, Current Trends and Future Perspectives. Springer-Verlag Berlin Heidelberg 2013. DOI: 10.1007/978-3-30595_14. pp. 337-348.
15. Maqsood, M.A., S. Hussain, **Tariq Aziz** and M. Ashraf. 2013. Sustainable Agriculture through Integrated Soil Fertility Management on Degraded Lands. In: "Developments in Soil Salinity Assessment and

Reclamation: Innovative Thinking and Use of Marginal Soil and Water Resources in Irrigated Agriculture" Shahid, S.A.; Abdelfattah, MA, Taha, FK. (Eds.) Springer.

RESEARCH ARTICLES IN PEER REVIEWED INTERNATIONAL JOURNALS

2025	
1	Wakeel, A., A. Qadeer, Z. Bano, M.R. Shahid, M. Rizwan, A. Kiran, M. Sanaullah, T. Aziz, R.M. Rees, A. Bhatia and J. Drewer. 2025. Managing Fertilizer Rates and Tillage Depth to Improve Nitrogen Use Efficiency and Soil Health. Journal of Soil Science and Plant Nutrition 25, 2937–2947 https://doi.org/10.1007/s42729-025-02310-5 Journal ISSN 0718-9508 Impact Factor 3.1 ISSN 0718-9516 Naqshe Zuhra, M. A. Maqsood, T. Aziz and M.I. Awan. 2025. Effect of feed and manure management on nitrogen footprints of commercial dairy farms: A survey study of Central Punjab. Soil Environ. 44(1): 104-113, 2025. https://doi.org/10.25252/SE/2025/253789 ISSN **2075-1141**
2024	
1	Rasheed, N., M.A. Maqsood, T. Aziz, M.I. Ashraf, I. Saleem, S. Ehsan, A. Nawaz, H.M. Bilal and M. Xu. Zinc portioning and allocation patterns among various tissues confers variations in Zn use efficiency and bioavailability in lentil genotypes. Frontiers Plant Sci. 14; 1325370, doi.org/10.3389/fpls.2023.1325370 **1664-462X Impact Factor 4.8**
2023	
1	Farooq, A., B. Saeed, H.M. Bilal, M.A. Maqsood, M. Nasim, T. Aziz and X. Mingang. 2023. Solublization and uptake of phosphorus (P) by wheat and maize crops from indigenous rock phosphate applied with biogas slurry and bagasse ash. Water, Air, Soil Pollution 234:709. 1573-2932
2	Möhring, N., Kanter, D., Aziz, T. et al. Successful implementation of global targets to reduce nutrient and pesticide pollution requires suitable indicators. Nat Ecol Evol (2023). 7, 1556–1559 https://lnkd.in/ezTchKcX **2397-334X**
3	Amin, S., T. Aziz, M. Z. Rehman, I. Saleem, M. Rizwan, A. Ashar, H. A. Mussawar, M.A. Maqsood. Zinc oxide nanoparticles coated urea enhances nitrogen efficiency and zinc bioavailability in wheat in alkaline calcareous soils. Environmental Science and Pollution Research 30: 70121-70130 https://doi.org/10.1007/s11356-023-27209-5 0944-1344

4	Ashraf, M.I., M.A. Maqsood, Tariq Aziz and M. F. Saleem. 2023. Elemental sulfur-assisted iron and zinc fortification for potential growth enhancement in canola under alkaline soil. Soil and Environment 42(2):223-235 10.25252/SE/2023/243255
5	Irfan, M. …… and Tariq Aziz*. Silicon Nutrition in Plants under Water-Deficit Conditions: Overview and Prospects. Water 2023, 15, 739. https://doi.org/10.3390/w15040739.
6	Ashraf, N., C. Hu, X. Xu, **Tariq Aziz**, L. Wu, M. A. Waqas, M. Farooq, X. Hu, W. Zhang, M. Xu. Long-term manure application increased soil organic carbon and nitrogen mineralization through the accumulation of unprotected and physically protected carbon fractions. **Pedosphere Volume 33, (2), 343-354** https://doi.org/10.1016/j.pedsph.2022.06.047 1002-0160
2022	
1	Abrar et al., 2022. Long-term manure application enhances organic carbon and nitrogen stocks in Mollisol subsoil. Land Degradation & Development. 34(3): 815-832 https://doi.org/10.1002/ldr.4498 1099-145X
2	Saeed, M.T. et al., 2022. Nutrient Accumulation During Vegetative and Reproductive Growth Affected by Endogenous and Exogenous Phosphorus Applications in Maize Crop. Commun. Soil Sci. Plant Analysis 54(7): 895-909. https://doi.org/10.1080/00103624.2022.2137180
3	Ehsan, S., M. A. Maqsood, A. Muhmood, G. W. Price and Tariq Aziz. Recycled biowaste improved soil physical properties of alkaline soil and wheat performance, Arabian Journal of Geosciences (2022) 15:1282 **https://doi.org/10.1007/s12517-022-10558-9.**
4	Hameed, M.K. et al., 2022. Differential Metabolic Responses of Lettuce Grown in Soil, Substrate and Hydroponic Cultivation Systems under NH4+/NO3− Application. Metabolites 12(5): 444 **DOI: 10.3390/metabo12050444**
5	Raghuram N, **Tariq Aziz** , S. Kant , J. Zhou and S. Schmidt. Editorial: Nitrogen Use Efficiency and Sustainable Nitrogen Management in Crop Plants. **Front. Plant Sci.** doi:

	10.3389/fpls.2022.862091
6	Sohail, M.I., M.Z. Rehman, **T. Aziz**, et al. Iron bio-fortification and heavy metal/(loid)s contamination in cereals: successes, issues, and challenges. **Crop & Pasture Science** https://doi.org/10.1071/CP21771
7	Bansal et al. Long-term trends of direct nitrous oxide emission from fuel combustion in South Asia". ***Environ. Res. Lett.*** 17 045028. 10.1088/1748-9326/ac5cf7
8	Umar, W., I. Czinkotaa, M. Gulyásb, **Tariq Aziz**, M. K. Hameed. Development and characterization of slow-release N and Zn fertilizer by coating urea with Zn fortified nano-bentonite and ZnO NPs using various binders. **Environmental Technology & Innovation** https://doi.org/10.1016/j.eti.2021.102250.
9	Saqib, M., A. Khaliq, A. Tanvir and **Tariq Aziz**. Impact of different nitrogen management options and weed competition periods on weed dynamics, productivity, and profitability of bread wheat. Arabian J Geosciences 15(9):865 10.1007/s12517-022-10146-x
2021	
1	Nawaz, A., M.A. Maqsood, M.I. Awan and **Tariq Aziz.** Comparative efficacy of natural and synthetic nitrification inhibitors for nitrogen release in alkaline calcareous soils. Pak J Agric Science 58(5), 1473-1482; DOI: 10.21162/PAKJAS/21. 58, 5(2021).
2	Ashraf et al., 2021. Soil microbial biomass and extracellular enzyme–mediated mineralization potentials of carbon and nitrogen under longterm fertilization (> 30 years) in a rice–rice cropping system. **Journal of Soils and Sediments 21: 3789-3800** https://doi.org/10.1007/s11368-021-03048-0.
3	Umar, W., M. K. Hameed, **Tariq Aziz**, et al. Synthesis, characterization, and application of ZnO nanoparticles for improved growth and Zn biofortification in maize, Archives of Agronomy and Soil Science, 67:9, 1164-1176, DOI: 10.1080/03650340.2020.1782893.
4	Ehsan, S., T. Aziz, M. A. Maqsood and A. Jabbar. 2021. Potential impact of bioenergy wastes as fertilizer supplement on soil properties. Pak. J. Agri. Sci., 58(3), pp. 891–904

5	Qudsia et al., 2021. Variation in growth, physiology, yield, and quality of wheat under the application of different zinc coated formulations. Appl. Sci. 2021, 11(11), 4797; https://doi.org/10.3390/app11114797
6	Fatima, S., M. Sabir, T. Aziz et al., 2021. Comparison of fine and coarse rice varieties for nickel accumulation and growth response at different levels of nickel. Clean - Soil, Air, Water 2021, 49(6), 2000336 https://doi.org/10.1002/clen.202000336
7	Saleem, I., M.A. Maqsood, T. Aziz, I.A. Bhatti and A. Jabbar. 2021. Potassium ferrite nano coated DAP fertilizer improves wheat (*Triticum aestivum* L.) growth and yield under alkaline calcareous soil conditions. Pak. J. Agric. Sciences. Pak. J. Agri. Sci., Vol. 58(2), 485-492; 2021
8	Saleem, I., M.A. Maqsood, Z. Rehman, **T. Aziz**, I.A. Bhatti and S. Ali. 2021. Potassium ferrite nanoparticles on DAP to formulate slow release fertilizer with auxiliary nutrients Ecotoxicology and Environmental Safety (215): 1, 112148 https://doi.org/10.1016/j.ecoenv.2021.112148
9	Qudsia et al., 2021. Efficiency of various formulations of urea coated with bioaugmented (Bacillus sp.) ZnO to improve growth, yield and Zn contents of wheat grains. **Pol. J. Environ. Stud**. Vol. 30, No. 1 (2021), 1-9.
10	Abrar, M.M., H. Xu, T. Aziz et al., 2021. Carbon, nitrogen, and phosphorus stoichiometry mediate sensitivity of carbon stabilization mechanisms along with surface layers of a Mollisol after long-term fertilization in Northeast China. Journal of Soils and Sediments **21,** 705–723 https://doi.org/10.1007/s11368-020-02825-7
2020	
1	Ashraf et al. 2020. Soil and microbial biomass stoichiometry regulate soil organic carbon and nitrogen mineralization in rice-wheat rotation subjected to long-term fertilization. **Journal of Soils and Sediments** 20, 3103–3113
2	Abrar, M., et al. Variations in profile distribution and protection mechanisms of organic carbon under long-term fertilization in a Chinese Mollisol. **Science Total Environment** 723: https://doi.org/10.1016/j.scitotenv.2020.138181 (10.75)

3	Irfan, M., **Aziz, T**., Maqsood, M.A. et al. Phosphorus (P) use efficiency in rice is linked to tissue-specific biomass and P allocation patterns. **Sci Reports (Nature)** 10, 4278 (2020). https://doi.org/10.1038/s41598-020-61147-3 (4.99)
4	Sabir, M., M. Z. Rehman, **T. Aziz**, H.R. Ahmad, Saifullah & E. A. Waraich. Comparative residual effect of activated carbon and other organic amendments on immobilization and phytoavailability nickel and other metals to Egyptian Clover (*Trifolium alexandrinum*) in contaminated soil. **Int. J. of Phyto-remediation**. 22(7): 687-693 DOI: 10.1080/15226514.2019.1707165
5	Ali, M., Maqsood, M. A., Aziz, T., & Awan, M. I. (2020). Neem (*Azadirachta indica*) oil coated urea improves nitrogen use efficiency and maize growth in an alkaline calcareous soil. *Pakistan Journal of Agricultural Sciences*, *57*(3), 675-684. https://doi.org/10.21162/PAKJAS/20.9850
6	Rashid, N., M.A. Maqsood, **T. Aziz**, et al. Characterizing lentil germplasm for zinc biofortification and high grain output. **J. Soil Sci. Plant Nutr.** 20, 336–1349 (2020)
2019	
1	Saeed, M.T, M. A. Wahid, M. F. Saleem, M. Shahid, T. Aziz, M. W. Ali. Improving phosphorous uptake efficiency and quality of maize through optimization of basal application. Maydica electronic publication – 2019 (Short communication).
2	Ashraf, M.N., T. Aziz, M.A. Maqsood, H.M. Bilal, …… M. Xu. Evaluating organic materials coating on urea as potential nitrification inhibitors for enhanced nitrogen recovery and growth of maize (*Zea mays* L.). Intl. J. Agric. Biology. IJAB-19-0824 (In Press).
3	Raza S., Z. Chen, M. Ahmed, M. R. Afzal, **T. Aziz** & J. Zhou. Dicyandiamide application improved nitrogen use efficiency and decreased nitrogen losses in wheat-maize crop rotation in Loess Plateau**. Archives of Agronomy & Soil Science.** 65(4) 450-464. 10.1080/03650340.2018.1506584.
4	Rasheed, N., M. A. Maqsood, T. Aziz, M. Z. Rehman, H. M. Bilal, M. A. Ayub, M. Irfan and M. Sanaullah. Zinc application methods affect its accumulation and allocation pattern in

	maize grown in solution culture. Intl. J. Agric. Biol. 21: 1197–1204
2018	
1	• Irfan, M., **T. Aziz,** M. A. Maqsood, H. M. Bilal, W.U. Khan and N. Rasheed. 2018. Differential Performance of Lowland Rice Cultivars for Phosphorus Uptake and Utilization Efficiency under Hydroponic and Soil Conditions. Intl. J. Agric. Biol. 21(3):703-710
2	• Mehmood, S., A. Jabbar, S.A.Anjum and T. Aziz, 2018. Assessment of bio-regulators to improve the performance of maize hybrids under spring conditions. Int. J. Agric. Biol., 20: 2491–2500.
3	Bilal, H.M., **T. Aziz**, M. A. Maqsood, M. Farooq and G. Yan. Categorization of Wheat Genotypes for Phosphorus Efficiency. **PLOS one** 13:10 https://doi.org/10.1371/journal.pone.0205471
4	Bilal, H.M., **T. Aziz**, M.A. Maqsood and M. Farooq. Grain phosphorus and phytate contents of wheat genotypes released during last 60 years and categorization of selected genotypes for phosphorus use efficiency. **Archives of Agronomy & Soil Science 65(6): 727-740** (ID: 1521957 DOI:10.1080/03650340.2018.1521957).
5	Khan, W.u.D., **T. Aziz**, M.A. Maqsood, M. Farooq, Y. Abdullah, P.M.A. Ramzani and H.M. Bilal. 2018. Silicon nutrition mitigates salinity stress in maize by modulating ion accumulation, photosynthesis, and antioxidants. **Photosynthetica**, 56: 1047-1057 https://link.springer.com/article/10.1007/s11099-018-0812-x.
6	Raza, S., J. Zhou, **T. Aziz**, R. Afzal, M. Ahmed; S. Javaid, Z. Chen. Piling up reactive nitrogen and declining nitrogen use efficiency in Pakistan: a challenge not challenged (1961-2013)" **Environmental Research Letters** 13(3): 034012 http://iopscience.iop.org/article/10.1088/1748-9326/aaa9c5/meta.
2017	

1	Saeed, M.T., M. A. Wahid, M. F. Saleem and **Tariq Aziz**. Enhancing phosphorus use efficiency by supplementing through soil applications and seed phosphorus reserves in maize (*zea mays*). Inter J. Agric. Biol. 19: 1394–1400.
2	• Khan, W., **Tariq Aziz**, M. Imran, P.M.A. Ramzani and T.G. Reichenauer. 2017. Silicon: a beneficial nutrient for maize crop to enhance photochemical efficiency of photosystem II under salt stress. **Arch. Agron. Soil Science**. 63(5): 599-611
2016	
1	Nazir, Q., M. Arshad, T. Aziz and M. Shahid. 2016. Influence of Zinc Impregnated Urea on Growth, Yield and Grain Zinc in Rice (*Oryza sativa*). Int. J. Agric. Biol., 18: 995–1000.
2	Maqsood, M.A., U.K. Awan, **T. Aziz,** H. Arshad, N. Ashraf and M. Ali. 2016. Nitrogen management in calcareous soils: problems and solutions. Pak. J. Agri. Sci., Vol. 51(1): 79-95.
• **2015**	
1	• Khan, W. **Tariq Aziz,** E.A. Warraich and M. Khalid. 2015. Silicon application improves germination and vegetative growth in maize grown under salt stress. Pak. J. Agri. Sci., Vol. 52(4): 937-944.
2	Khan, W, M. Faheem, M. Y. Khan, S. Hussain, M. A. Maqsood and **Tariq Aziz.** 2015. Zinc requirement for optimum grain yield and zinc biofortification depends on phosphorus application to wheat cultivars. Romanian Agricultural Research 32. 1-9.
3	Nasim, M., Z. Rengel, **Tariq Aziz,** B.D. Regmi. 2015. Boron toxicity alleviation by zinc application in two barley cultivars varying in tolerance to boron toxicity. Pakistan J. Agri., Sci..52: 151-158
4	**Aziz, T.,** H. Lambers, D. Nicol and M. H. Ryan. 2015. Mechanisms for tolerance of very high tissue phosphorus concentrations in *Ptilotus polystachyus*. **Plant Cell and Environment** 38, 790–799. (Impact Factor: 6.97).
	Maqsood, MA, S. Hussain, M. Ahmad, **Tariq Aziz**, M.A. Naeem, H.A. Raza and M. Hussain. 2015.

5	Zinc indexing in wheat grains and associated soils of southern Punjab. Pak. J. Agric. Sci. 52(2), 429-436
6	Aziz, H., M. Sabir, H.R. Ahmad, **T. Aziz**, M. Z. Rehman, K. R. Hakeem, M. Ozturk. 2015. Alleviating effect of calcium on nickel toxicity in rice. **Clean- Soil Air Water**. doi: [10.1002/clen.201400085] 43: 901-909
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3	**Aziz, T**., P.M. Finnigen, H. Lambers and R. Jost. 2014. Organ specific phosphorus allocation patterns and transcript profiles linked to phosphorus efficiency in two contrasting wheat genotypes. **Plant Cell Environment**. 37: 943-960. (Impact Factor: 5.91).
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6	Sabir, M., M.M. Hanafi, M.Z. Rehman, Saifullah, H.R. Ahmad, K.R. Hakeem, W. Gandahi and **Tariq Aziz**. 2014. Comparison of low molecular weight organic acids and ethylene diamine tetra acetic acid to enhance phyto-extraction of heavy metals by maize (*Zea mays* L.) in contaminated Soil. **Communications in Soil Sci Plant Anal**. 45:42–52, 2014**.**
2013	
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4	Sabir, M., MM Hanafi, **Tariq Aziz**, HR Ahmad, MZ Rehman, Saifullah, G. Murtaza and KR Hakeem. 2013. Comparative effect of activated carbon, press-mud and poultry manure on immobilization and concentration of metals in maize (*Zea mays* L.) grown on contaminated soil. Int. J. Agric & Biol. 15:559-564.
5	Hussain, S., M.A. Maqsood, **Tariq Aziz** and S.M.A. Basra. 2013. Zinc bioavailability response curvature in wheat grains under incremental zinc applications. **Archives of Agronomy and Soil Science** 59(7): 1001-1016.
2012	
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4	Waraich, E.A., R. Ahmad, A. Haleem and **Tariq Aziz**. 2012. Alleviation of temperature stress by nutrient management in crop plants: a review. **J. Soil Sci. Plant Nutr** 12 (2), 225-248.
5	Jaafer, Z.M., M. Farooq, M.A. Cheema, I. Afzal, S.M.A. Basra, M.A. Wahid, **Tariq Aziz** and M. Shahid. 2012. Improving the performance of wheat by seed priming under saline conditions. **Journal of Agronomy and Crop Science** 198 (1): 38-45.
6	Tahir, M.A., **Tariq Aziz**, M. Farooq and G. Sarwar. 2012. Silicon induced changes in growth, ionic composition, water relations, chlorophyll contents and membrane permeability in two salt stressed wheat genotypes. **Archives of Agron. Soil Sci 58(3):247-256.**
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2	Ahmad, W., Rahmatullah, M. A. Gill and **Tariq Aziz**. 2006. Bioavailability of inorganic phosphorus fractions in calcareous soils estimated by Neubauer technique, iron impregnated filter paper and

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3	Ahmad, W., Rahmatullah, and **Tariq Aziz**. 2006. Phosphorus desorption/release kinetics from different depths of calcareous soils by iron-coated filter paper strips. Journal of Botany 1(1): 117-123.
4	Ali, L., Rahmatullah, A.M. Ranjha, **Tariq Aziz**, M. A. Maqsood. And M. Ashraf. 2006. Differential potassium requirements and its substitution with sodium in different cotton cultivars. Journal of Agricultural Sciences 43 (3-4): 108-113.
2005	
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2	**Tariq Aziz**, Rahmatullah, M. Aamer Maqsood and T. Mansoor. 2005. Differences in phosphorus absorption, transport and utilization by twenty rice (*Oriza sativa* L.) cultivars. Journal of Agricultural Sciences 42 (3-4): 8-13
3	Gill, M. A., M. Sabir, Rahmatullah and **Tariq Aziz**. 2005. Effect of phosphorus stress on growth, phosphorus uptake and utilization efficiency of different cotton cultivars. Pakistan Journal of Agricultural Sciences 42(1-2): 42-47.
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2	Gill, M.A., F. Ahmad, **Tariq Aziz**, Rahmatullah and M. A. Tahir. 2004. Growth and phosphorus uptake by *Brassica* cultivars grown with adequate and deficient phosphorus levels. Pakistan Journal of Agricultural Sciences 41(3-4): 114-118.
3	Gill, M. A., , **Tariq Aziz**, and Rahmatullah. 2004. Differences in phosphorus-zinc interaction among sunflower (*Helianthus annus* L.), Brassica (*Brassica napus* L.) and Maize (*Zea mays* L.). Journal of

	Agricultural Sciences 41 (1-2): 29-34.
2003	
1	Tahir, M. A., Rahmatullah, M. A. Gill, **Tariq Aziz** and M. Imran. 2003. Response of wheat and oat crops to potassium application and artificial irrigation with canal water. Journal of Agricultural Sciences 40 (3-4): 114-118.
2	Sabir, M., M. A. Gill, Rahmatullah, **Tariq Aziz** and M. A. Maqsood. 2003. Differences among rice cultivars in potassium uptake and its utilization. Journal of Agricultural Sciences 40 (3-4): 119-121.
3	Kosar, H. S., M. A. Gill, **Tariq Aziz,** Rahmatullah and M.A. Tahir. 2003. Relative phosphorus utilization efficiency of wheat genotypes in hydroponics. Journal of Agricultural Sciences 40 (1-2): 28-32.
4	Imran, M., Rahmatullah, M.A. Gill and **Tariq Aziz.** 2003. Differential P mobilization by two wheat genotypes on a calcareous soil fertilized with ammonium sulfate, potassium nitrate, ammonium nitrate and urea. Journal of Soil Science, 22(4): 64-67
2002	
1	Kosar, H. S., M. A. Gill, Rahmatullah, **Tariq Aziz** and M. Imran. 2002. Solubilization of tri-calcium phosphate by different wheat genotypes. Journal of Agricultural Sciences 39(4): 273-277
2	**Tariq Aziz**, M. A. Gill and Rahmatullah. 2002. Silicon nutrition and crop production: **A review**. Pakistan Journal of Agricultural Sciences 39 (3): 181-187
3	Akhtar, M. S., M. A. Gill, **Tariq Aziz** and Rahmatullah. 2002. Differential phosphorus requirement and utilization efficiency of *Brassica* genotypes. Journal of Agricultural Sciences 39 (3): 188-192
4	Gill, M. A., , **Tariq Aziz,** Rahmatullah and M. S. Akhtar. 2002. Differential growth response and phosphorus utilization efficiency of rice genotypes. Journal of Agricultural Sciences 39 (2): 83-87

5	Rahmatullah, M. Salim, M. A. Gill, K. Hussain and **Tariq Aziz**. 2002. Soil burial influencing response to applied phosphorus in a torrifluvent. Journal of Agricultural Sciences 39 (2): 104-108
2001	
1	**Tariq Aziz**, M.A. Gill and Ahmed. 2001. Differential growth response of cotton genotypes infected with root rot to silicon nutrition. Pakistan Journal of Soil Science 20: 101-108.

Organizers/member of Scientific Events

1. Organized a Consultative workshop on "UNEA-7 Resolution draft on Sustainable Nitrogen management" 4th July, 2025, Marriot Hotel, Islamabad, collaborated by MoCC&EC and UAF.
2. Organized "Sustainable Nitrogen Management: A Way Forward for Climate, the Environment, and Food Security". GIZ Bonn, Germany. Friedrich-Ebert-Allee 36, 53113 Bonn, 16 June, 2025, organized by UAF, CIWF, INI, INMS and UNEP
3. Organizer, World Soil Day Celebrations, Seminar and Student Competition, 20 March 2025, UAF Constituent College, Depalpur, Okara.
4. Focal Person, Organizing Committee, Seminar and Nitrogen Policy Report Launching Ceremony, 5-6 August 2024, under GCRF-South Asia Nitrogen Hub, at CAS Auditorium UAF.
5. Member of organizing Committee in a 3 days workshop on "Modelling nitrogen management in different cropping systems in changing climate" from 20th to 22nd May at CAS Auditorium and ISES, UAF.
6. Focal person of a consultative workshop on "South Asia Roadmap towards Sustainable Nitrogen management" in collaboration with MoCC and COMSATS University Islamabad, organized under GCRF-South Asia Nitrogen Hub, at Marriot Hotel, Islamabad from 8th to 9th May, 2024.
7. Member of International Committee of 9th International Nitrogen Conference "Balancing Global Change and Sustainability" Feb 5-8, 2024 at GGSI University, India.
8. Organizer, Side Event hosted by Ministry of Climate Change and Environmental Coordination, Govt of Pakistan at COP29, Baku, Azerbaijan, 20th November 2024.
9. Member of Organizing Committee, Side Event at UNEA-6 "Making Nitrogen Visible through SDGs"
10. Member of Organizing Committee "Side Event at UNFCC Bonn, 2024, "Making Nitrogen Visible through Rio Conventions".
11. Member Organizing Committee, Ist International Conference on Soil Pollution and Remediation, March 15-16, 2023. FC College University, Lahore, by Department of Environmental Sciences
12. Member, Organizing Committee, International Conference Climate Change, 2022. Institute of Soil & Environmental Sciences, University of Agriculture, Faisalabad.
13. Member, Organizing Committee, International Wheat Conference, October 19-20, 2022, University of Agriculture, Faisalabad.

SCIENTIFIC CONFERENCES/SEMINARS/SYMPOSIA/WORKSHOPS

YEAR	CONFERENCES/WORKSHOPS
2025	*Participated in the 7th United Nations Environment Assembly from 9th to 12th December, 2025 at Nairobi, Kenya*
2025	*Participated in Annual meeting of International Nitrogen Network Project of INI from 10th to 12th December, 2025.*
2025	*Invited Speaker at 7th Nitrogen Working Group of United Nations Environment Group (Online) from 1-3 July 2025.*
2025	*Invited expert speaker at "Sustainable Nitrogen Management: A Way Forward for Climate, the Environment, and Food Security". GIZ Bonn Friedrich-Ebert-Allee 36, 53113 Bonn, 16 June, 2025, organized by UAF, CIWF, INI, INMS and UNEP*
2025	*Invited Speaker at 2nd International Conference on Climate Change Impacts on Agriculture and Food Security, organized by University of Swabi, 22-24th April, 2025. The title of talk "Sustainable Nitrogen Management for Food Security, Climate Action and Environmental Sustainability.*
2025	*Invited as expert/discussant on a Workshop "Climate Resilient and low carbon agriculture in Pakistan" organized by Asian Development Bank, on 11th February 2025, Flatties Hotel, Lahore, Pakistan.*
2024	*Invited talk on "Side Event "Sustainable Nitrogen Management; Collaborative solutions for Climate Change, Environment and Food Security" at COP29 on 20th November, 2024, Baku Azerbaijan.*
2024	*Invited talk on "Side event of HLPF 2024 on Making Nitrogen Visible; Reflection through SDGs under Review" Online, organized by CIWF, INI and UAF…. Organizer*
2024	*Invited Talk on "Draft South Asia Roadmap for Sustainable Nitrogen Management" in 6th*

	Nitrogen Working Group Meeting of UNEP, 27 June 2024, Online
2024	*Organized and talk at official side event "Making Nitrogen Visible through Sustainable Development Goals" at UNEA-6, Nairobi, Kenya, 26 February, 2024….. Organized*
2023	*Invited Talk on "Status of Reactive Nitrogen Pollution in South Asia" at a Symposium organized by Northwest A&F University, Yangling, Shaanxi, China, from 13 to 15 December, 2023*
2023	*Invited Talk on a side event "Grasping Nitrogen" along 5th Forum of Ministers and Environment Authorities of Asia Pacific, organized by MoE, Sri Lanka, SANH and SACEP, at Shangrilla Hotel, Sri Lanka on 3rd October, 2023.*
2023	*Invited Talk on 4th meeting of UNEP-Working Group on Nitrogen (Online), 29th September, 2023, on "South Asia Roadmap for Sustainable Nitrogen Management".*
2023	*Organized and Plenary Talk on "Zero Draft of UNEA-6 Resolution on Sustainable Nitrogen Management and Roadmap for SNM" in collaboration with Ministry of Climate Change & Environmental Coordination, SANH, INI, and UAF on 22 September, 2023 at Marriot Hotel, Islamabad.*
2023	*Organized and Plenary Talk on "Roadmap to Sustainable Nitrogen Management" in Stakeholders Engagement Event organized by UAF in collaboration of SAU at Hotel MovenPick, Karachi, 18th July, 2023.*
2023	*Invited Talk in "International Workshop on Reactive Nitrogen Management" organized by Bangabandhu Sheikh Mujibur Rahman Agricultural University, 22 March 2023. (Remotely*
2023	*Invited Talk in "SANH (India) Stakeholders' Workshop on Sustainable Nitrogen Management -2023" August 25-26, 2023" Kalinga Institute of Industrial Technology, Bhubaneswar, India (Remotely)*
2023	*Invited Talk at Ist International Conference on Soil Pollution and remediation 15-16 March, 2023 at FCCU, Lahore "Food Security and Environment Degradation; Nitrogen on the Driver's seat".*

2023	*2nd International Conference "New Trends in Biological Sciences" University of Okara, Okara January 16-17, 2023.*
2022	*21st Nitrogen Workshop from 24th to 26th October 2022 at Madrid, Spain (Joined remotely and presented an oral talk)*
2022	*Soil Science Society of Pakistan Congress, University of Agriculture, Faisalabad; Presented invited speaker: "Why care about nitrogen",*
2022	*National Conference On Soil Degradation: An Alarming Threat To Food Security And Environment Organized by Department of Soil Science, Faculty of Agricultural Sciences & Technology, Bahauddin Zakariya University, Multan, Pakistan, 3-4 March 2022, Invited Talk on "Evaluation of natural and synthetic nitrification inhibitors on wheat yield and nitrogen use efficiency".*
2021	*Sustainable N management for food security, organized by UAF, FFC, SANH, PAS, International Webinar on 8th December, 2021. Invited Speaker "Pakistan Nitrogen Assessment".*
2021	*International Symposium on Plateau Ecological Environment Protection and High-quality Development of the Yellow River Basin. October 30, 2021, (Webinar) organized by College of Natural Resources and Environment, Northwest A&F University, Yangling, Shaanxi, China. Presented an oral talk "Sustainable Nitrogen Management for Food Security and Environment"*
2021	*Sustainable N management for ecosystem restoration, International World Environment Day Webinar 4th June, UNEP, SANH, UAF. Organizer and Speaker.*
2021	*Food security and environmental concerns of reactive nitrogen; Pakistan's perspective. 13th September 2021, ICESD, SDSC, GC University, Lahore, Oral Presentation and Session Chair*
2020	*C3 Demo Meeting, INA (INMS) Online meeting 23.11.2020, UK.*

2020	*INA Book Meeting Online: 30 Nov to 2 December and 8 December to 10 December*
2020	*Nitrogen Challenges in Agri-Food Systems: Halve Nitrogen Waste by 2030, Organized by FAO (Online Webinar) December 07, 2020*
2020	*Covid 19, Plenary Meeting, 28th November, 2020 (ONLINE) SANH Project* *Work Package Meetings (Online) Covid-19 Impact. SANH 5-6, 8-10 November 2020*
2020	*INMS Plenary Meeting (International) ONLINE MEETING with more than 100 participants*
2019	*UN Nitrogen Awareness Campaign, Sri Lanka*
2019	*Plenary Meeting INMS-4 (International Nitrogen Mangement System), 28 April to 03 May, 2019, UN Environment Base, Nairobi, Kenya*
2019	*Inception Meeting of South Asia Nitrogen Hub at Khatmandu, Nepal, 24 to 28 February, 2019*
2019	*Concluding Workshop of a German Project "Assessing the biomass potential for ensuring the soil health and food security in Pakistan" 17-20 Feb 2019.*
2019	*Research Visit of ITT, University of Cologne, Germany (January 10 to January 25, 2019)*
2018	*INMS-3, International Nitrogen Management System-3, Universty of Edinberg, UK, 21-25 April, 2018*
2017	*Kick-off meeting of a german funded project at IATT, University of Cologn, Germany, December 14 to 18, 2017*
2017	*XVIII International Plant Nutrition Colloquium 19-24 August 2017 • Copenhagen |*

	Denmark
2016	*IUFRO Regional Congress on Asia and Oceania 2016, 24-27 October, Beijing (as invited speaker) China National Convention Centre, Beijing, China.*
2016	*7th International Crop Science Congress organized by Institute of Crop Science, Chinese Academy of Agricultural Sciences, August 14-19, 2016, International Conventional Centre, Beijing, China (as Speaker).*
2015	*Workshop on Applications of Biotechnology for Wheat Improvement organized by National Institute of Biotechnology and Genetic Engineering, (NIBGE), Faisalabad.*
2015	*International Workshop on Current Status of Fertilizer Use in Pakistan, organized by Institute of Soil & Environmental Sciences and ICARDA, Nov. 29 – Dec. 01, 2015, at UAF (as Invited Speaker).*
2015	*International Conference on Soil Sustainability and Food Security organized by Institute of Soil & Environmental Sciences, UAF, 15-17 November, 2015 at New Senate Hall, UAF.*
2015	*Workshop on "Balanced Use of Fertilizers in Pakistan" by International Potash research Institute & Institute of Soil & Environ. Sci, UAF, 17 March, 2015 at STC, UAF. (as a Resource Person)*
2014	*Indigenous on-campus capacity sharing workshop organized by ORIC, UAF*
2014	*Capacity Building Seminar on "Balanced Use of Fertilizers in Pakistan" organized by International Potash Institute & Inst. Soil & Environ. Sciences, UAF, 20th August, 2014 at Serena Hotel, Faisalabad.*
2013	*Project Formulation Workshop, December 3-4, 2013, organized by Pakistan Science Foundation and University of Agriculture, Faisalabad.*
2011	*International Conference "Rhizosphere3" at Perth, Western Australia, by The University*

	of Western Australia, 25-30 September, 2011.
2011	*Workshop "Phosphorus – The inside Story" 02 Feb. Organized by Faculty of Natural and Agricultural Sciences and UWA Institute of Agriculture, at School of Plant Biology, University of Western Australia, Crawley, WA 6009, Australia.*
2006	*First Joint meeting of Plant Breeders and German Society of Plant Nutritionists "Plant Breeding Meets Plant Nutrition" held at of , on 26-28th September, 2006.*
2005	*Annual Conference of German Plant Nutrition Society, 27-28 September, 2005. .*

Advisory Services

1. Organizer of "Potato Festival" in collaboration with Potato Growers Coop Society and Potato Research Institute, Sahiwal at Sub-Campus from 21-23 March, 2022. Coord/7-D/1407-30, dated 28.2.2022.
2. Focal person of Wheat Campaign 2023 for Sub-Campus Depalpur, Okara, vide letter No. SCDO/2020, Dated 20.10.2023.